

Unique LED LANTERN

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Presented here is an LED lantern that also functions as an emergency light or a tent light. The design has all required functions, including on/off switch, maintenance-free rechargeable battery, integrated battery charger and more. The built-

in battery can be recharged using any 500mA external DC input voltage ranging from 9V to 18V using its onboard voltage regulator.

Circuit and working

The circuit diagram of the unique LED

charger is realised without using a dedicated SMD chip, such as TP4056. Instead, two small signal transistors S8550 and S8050 (T1 and T2) and one precision adjustable shunt regulator chip TL431 (IC2) are used.

The 6V relay (RL1) functions as an automatic light switch. When no external DC supply is available at connector CON1, RL1 is de-energised and its common pole is connected to NC contact. When switch S1 is closed, LED3 gets 3.7V supply from back-up battery BATT.1 through a resistor (R8), which sets the current within safe limits. Close S1 to turn on LED3.

When external DC supply is available at CON1, RL1 is energised and its pole is connected to NO terminal. During this time, LED3 will be off, irrespective of S1's position. Charging voltage is available across BATT.1.

BATT.1 is a single-cell (1S) lithium-ion type with 3.7V nominal voltage and 4.2V maximum voltage. Any compact cellphone battery can be

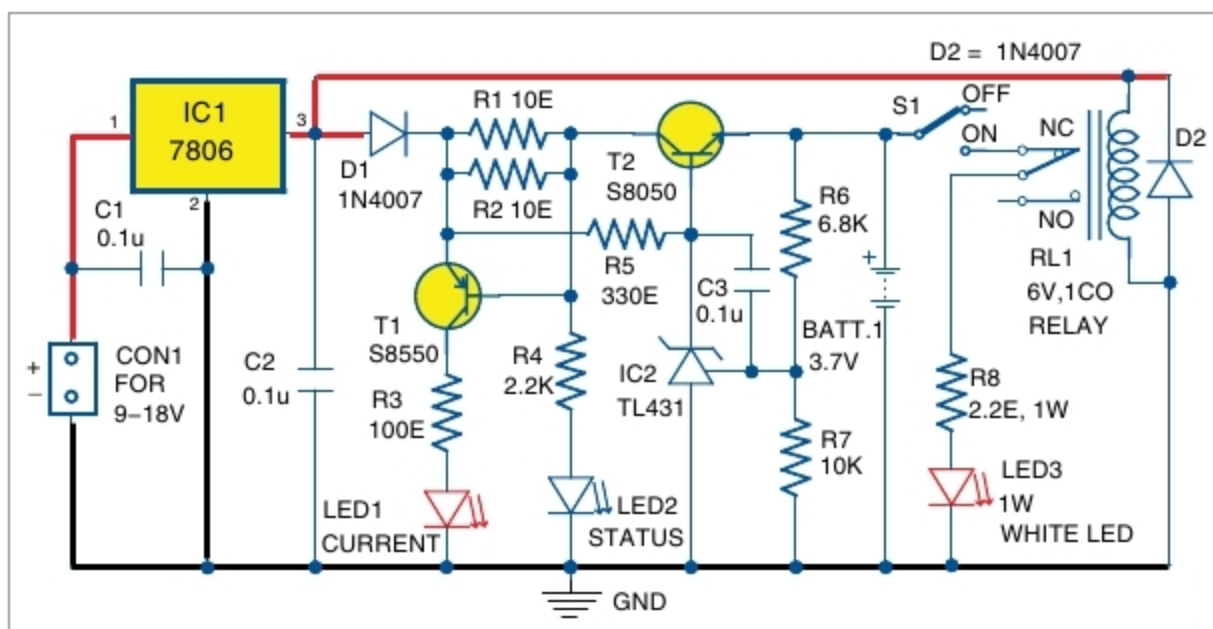


Fig. 1: Circuit diagram of unique LED lantern

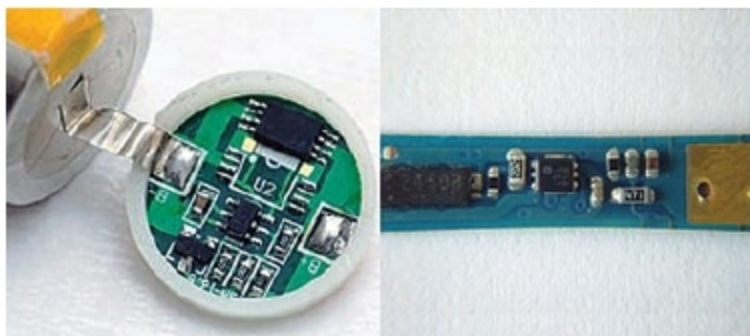


Fig. 2: Protection board inside 18650 lithium-ion cell (left) and Nokia BL-5Cx battery (right)

lantern is shown in Fig. 1. It has only a few inexpensive and discrete components. Front-end of the circuit is a pre-regulator, wired around popular 3-pin fixed-voltage regulator 7806 (IC1) for converting 9V DC - 18V DC supply to a stable 6V DC output. Here, the battery

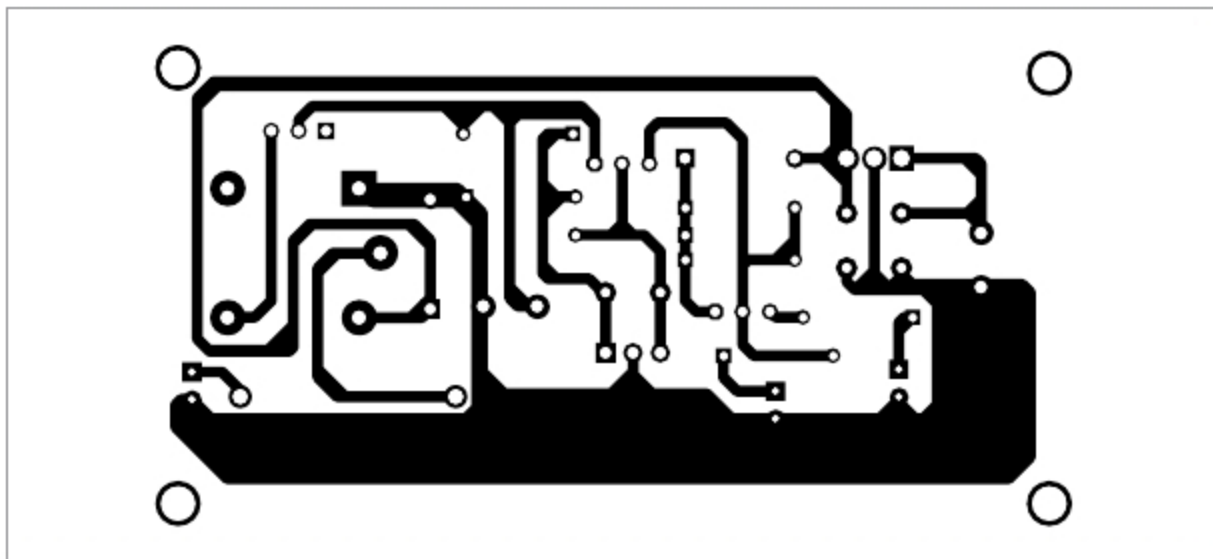


Fig. 3: Actual-size PCB layout of unique LED lantern

PARTS LIST

Semiconductors:

IC1	- 7806, 6V voltage regulator
IC2	- TL431 shunt regulator
T1	- S8550 pnp transistor
T2	- S8050 npn transistor
D1-D2	- 1N4007 rectifier diode
LED1, LED2	- 3mm LED
LED3	- 1-watt white LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

R1, R2	- 10-ohm
R3	- 100-ohm
R4	- 2.2-kilo-ohm
R5	- 330-ohm
R6	- 6.8-kilo-ohm
R7	- 10-kilo-ohm
R8	- 2.2-ohm, 1-watt

Capacitors:

C1-C3 - 0.1uF ceramic disc

Miscellaneous:

CON1	- 2-pin terminal connector
S1	- On/off switch
RL1	- 6V single-changeover sugar cube relay
BATT.1	- 3.7V rechargeable lithium-ion battery
	- 9V-18V DC power supply